GridPulse Project Study Notes

A beginner guide to predictive and prescriptive analytics for cross-border power grids

Designed to be studied in approximately 30 minutes

This guide explains the complete project in plain language. You do not need prior knowledge of artificial intelligence, machine learning, power systems, or predictive analytics. By the end, you should be able to explain what the project receives, what it calculates, how it makes a forecast, how it detects stress, how it proposes a reroute, and how the models are evaluated.

The one sentence answer

GridPulse reads hourly electricity loads and cross-border flows, turns them into a clean network time series, predicts the next hour's flow for each country, labels possible stress, and ranks a safer alternate route using data quality, headroom, distance, and estimated loss.

Thirty minute reading plan

Minutes	Read this section	What you should know afterwards
0-3	The problem in plain language	What a node, link, load, flow, and stress mean
3-7	The data	Why there are two CSV files and what they contain
7-12	End-to-end workflow	How raw files become a model input
12-16	Feature extraction	Why 11 features are created
16-22	Models and training	How baselines, Fusion GRU, and DCRNN learn
22-26	Risk, rerouting, and loss	How a prediction becomes a recommendation
26-29	Results and evaluation	What the metrics really say
29-30	Recap and viva questions	How to explain the project confidently

Important conclusion: this is a decision-support system and a research benchmark. It does not switch equipment, dispatch generators, or claim engineering-grade line losses because the supplied data does not contain surveyed line parameters.

1 The problem in plain language *(3 minutes)*

Electricity must be balanced every hour. If demand rises in one country, generation falls in another, or a transfer corridor becomes heavily used, the pattern of power moving between countries changes. An operator needs two answers: what is likely to happen next, and what action should be investigated before the situation becomes unsafe.

The basic vocabulary

Term	Plain meaning	How GridPulse uses it
Node	A country or grid area represented as a point in the network.	The 12 countries are the nodes.
Edge or link	A connection through which power can move between two nodes.	The canonical graph contains 24 links.
Load	Electricity being consumed at a time.	Load pressure helps explain future transfers.
Flow	Power moving from one country to another.	The target is each country's net flow in MW.
Net flow	A signed summary of incoming and outgoing transfer.	The sign tells direction; magnitude tells scale.
Stress	A screening label showing unusually large flow magnitude.	Normal, Warning, or Critical.
Forecast	A numerical estimate of what comes next.	Next-hour net flow for all 12 nodes.
Recommendation	A suggested action for a human to investigate.	A possible alternate route with checks.

Predictive versus prescriptive analytics

Predictive analytics answers: What is likely to happen? In this project, the model estimates next-hour net flow. Prescriptive analytics answers: What could we do about it? In this project, the routing layer searches connected alternate paths, rejects paths with insufficient headroom or critical links, estimates the change in proxy loss, and ranks the remaining options. The recommendation is not an automatic command.

A simple analogy is weather forecasting. Predictive analytics says there is a high chance of rain. Prescriptive analytics says carry an umbrella and avoid the flooded road. GridPulse applies the same separation to power-flow monitoring.

2 What the data contains *(4 minutes)*

The project uses two complementary files. The first describes what each country is experiencing over time. The second describes how countries are connected by interconnectors. Using both makes it possible to combine time-series behavior with network structure.

File	What one row means	Important fields	Role
dynamic_with_flows.csv	One country at one hourly timestamp	datetime, country, load, net_flow, imbalance, residual, centrality	Time-varying measurements and engineered signals
power_flow.csv	One directional transfer record at one timestamp	datetime, interconnector, source, target, power_flow_mw	Physical connection and transfer evidence

Dataset facts from this run

Item	Observed value	Why it matters
Dynamic data	315,598 rows and 37 columns	Large hourly country-level history
Power-flow data	1,262,592 rows and 5 columns	Directional interconnector records
Network	12 countries and 24 canonical links	The graph is small enough to audit visually
Period	2022-01-01 to 2024-12-31	Historical study window
Aligned history	960 hourly points per country	Common input length after alignment
Data checks	0 missing cells and 0 duplicate keys in both source tables	No obvious null or duplicate-row defect
Gaps and aliases	38 country timestamp gaps; 7 aliases normalized	Quality signals are retained, not hidden

Why cleaning is essential

1. Parse timestamps as UTC so the same hour means the same thing in every country.
2. Check missing cells, duplicate keys, invalid country codes, and hourly gaps.
3. Canonicalize interconnector names so several labels for the same endpoint pair do not create fake edges.
4. Align the common recent history so every node has comparable time positions.

Source in project: results/dataset_profile.json and results/preprocessing_summary.json

3 The complete end-to-end workflow *(5 minutes)*

The workflow is a chain. Each stage has one job and produces an artifact that the next stage can inspect. This is important because a prediction is only trustworthy when the path from raw data to result is understandable.

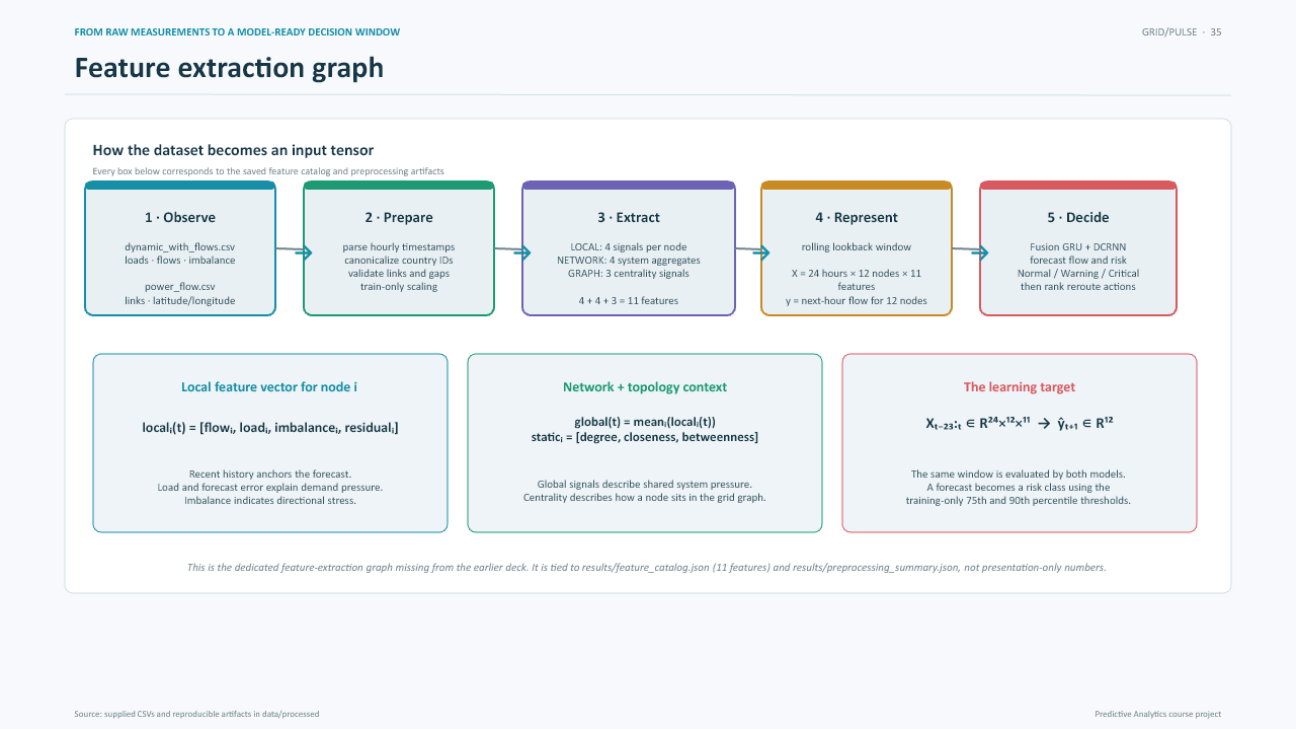


Figure 1. The project workflow from the two CSV files to the model decision.

Stage	What happens	Output
1 Observe	Read country measurements and interconnector records.	Raw dynamic and network tables
2 Prepare	Parse, validate, canonicalize, align, and scale.	Clean hourly node and edge data
3 Extract	Create local, network, and static topology features.	11 numerical features per node and hour
4 Represent	Build rolling sequences of the previous 24 hours.	X window with shape 24 x 12 x 11
5 Predict	Fusion GRU and DCRNN estimate next-hour flows.	12 predicted net-flow values
6 Classify	Compare absolute flow magnitude with training thresholds.	Normal, Warning, or Critical
7 Prescribe	Check headroom, loss proxy, and alternate paths.	Ranked recommendation for review
8 Display	Serve the result through API endpoints and the dashboard.	Human-readable live replay and charts

The browser's live view is a historical replay of the supplied timeline, not a connection to a utility control center. The API shape is ready for a future telemetry connector, but the current evidence comes from the saved dataset.

4 Feature extraction in simple terms *(4 minutes)*

A model cannot reason directly from a spreadsheet row in the same way a person does. Feature extraction converts raw fields into a compact set of numerical signals that describe recent behavior, system-wide context, and where each country sits in the network.

Feature group	Count	Examples	Why it helps

Feature group	Count	Examples	Why it helps
Local time series	4	Local net flow, local load, local imbalance, local load residual	Describes what is happening at that country now
Network context	4	Global mean net flow, global mean load, global imbalance, global residual	Describes shared system pressure
Static graph	3	Degree, closeness, betweenness centrality	Describes connectivity and bridge-like importance
Total	11	4 + 4 + 3	The model input feature count

The most important calculations

$$\text{local_i}(t) = [\text{flow_i}(t), \text{load_i}(t), \text{imbalance_i}(t), \text{residual_i}(t)]$$

A four-number summary of node i at hour t

$$\text{global}(t) = \text{mean over all nodes of local signals}$$

A compact view of the wider system

$$\text{X window} = 24 \text{ hours x } 12 \text{ nodes x } 11 \text{ features}$$

The model receives one day of history for every country

The target is the next-hour net flow vector for all 12 nodes. In other words, the model reads 24 hourly snapshots and predicts the next snapshot. The feature catalog is saved so a reviewer can see the name, source, formula, and reason for every feature.

What the feature study found

Feature	Permutation MAE increase	Ablation MAE change
local net flow	+3733.1 MW	+1993.3 MW
local load	+180.7 MW	-0.8 MW
local imbalance	+107.8 MW	+20.9 MW
betweenness centrality	+97.2 MW	+35.2 MW
closeness centrality	+70.7 MW	+17.1 MW
degree centrality	+50.4 MW	+19.0 MW

Local net flow is the strongest signal in the permutation study: if its values are shuffled, validation error rises sharply. This is sensible because recent flow is an autoregressive anchor. Centrality features also help, but the current run does not prove that every global feature is useful; some have near-zero or negative ablation changes. That is exactly why ablation studies are included.

Source in project: results/feature_catalog.json and results/feature_importance.json

5 The prediction models *(6 minutes)*

The project does not trust one algorithm automatically. It compares simple baselines with two trained neural models. A baseline is a reference point: if a complicated model cannot beat a simple rule, the extra complexity is not justified for this dataset.

Model	Beginner explanation	What it tests
Persistence	Predict that the next value will be the most recent value.	Is the flow so stable that no learning is needed?
24 hour moving average	Average the previous 24 hours.	Does smoothing remove noise and help?
Seasonal naive	Use the value from the same hour yesterday.	Is a daily repeating pattern enough?
Linear trend	Fit a straight line to recent values and extend it.	Is a simple upward or downward trend enough?
Fusion GRU	A gated memory network reads the 24-hour sequence plus context features.	Can learned time patterns improve the forecast?
DCRNN	A recurrent model that also diffuses information across connected countries.	Does graph neighborhood context improve the forecast?

How Fusion GRU works

A GRU is a recurrent neural network with gates. Think of it as a notebook that reads one hour at a time. At each hour it decides what old information to keep, what new information to add, and what information to pass to the next hour. The final hidden state summarizes the 24-hour window, and a final layer converts that summary into 12 next-hour flow predictions.

$$\text{hidden state at time t} = \text{GRU}(\text{current features at t, previous hidden state})$$

The memory is updated one hourly snapshot at a time

How DCRNN works

DCRNN keeps the country axis instead of flattening the grid into one list. At each recurrent step, it applies two-step forward and reverse diffusion over the canonical graph. Forward diffusion asks what information can arrive along outgoing connections. Reverse diffusion asks what information can arrive from the opposite direction. The recurrent update then uses that graph-aware information.

In this project DCRNN uses the same 24-hour lookback, training-only scaling, chronological split, and target as Fusion GRU. That makes the comparison fair. It does not solve a full power-flow equation and it does not automatically know voltage, thermal limits, or dispatch constraints.

Why DCRNN can lose even though it knows the graph

- The graph has only 12 nodes and 24 links, so global features may already capture much of the useful shared context.
- The data does not contain surveyed line geometry, ratings, voltage, or dispatch constraints, so the graph is informative but incomplete.
- Graph diffusion adds parameters and computation. If the extra information is weak, the model can overfit or add noise.
- A weaker DCRNN score is still valuable evidence: it tells us not to claim that graph structure always wins.

6 How training and validation work *(4 minutes)*

Training means showing historical examples to a model and adjusting its internal weights so its predictions become closer to the observed next-hour flows. Validation is a separate time period used to monitor learning and tune choices. The final test holdout is kept for the honest end check.

The leakage-safe split

Partition	Samples	Purpose
Training	1,800	Fit model weights
Validation	504	Monitor generalization and compare choices
Test holdout	2,016	Final chronological evaluation

The model looks back 24 hours and moves the window by 4 hours. The final test horizon is 168 hours. Scaling means converting large and small fields to comparable ranges, but the mean and standard deviation are fitted only on the training window. If test information were used to scale training, the result would leak future knowledge into the past.

What the training curves mean

- Training loss falling means the model is fitting the examples it sees.
- Validation loss falling with training loss means the model is learning patterns that transfer to unseen time periods.
- Training loss falling while validation loss rises suggests overfitting: the model memorizes the training period.
- Both curves staying high suggests underfitting, weak features, insufficient training, or a difficult dataset.

$$\text{MSE} = \text{average of (actual - predicted)}\text{^}2;\ \text{RMSE} = \text{square root of MSE}$$

The squared error penalizes large mistakes more strongly

$$\text{MAE} = \text{average of absolute(actual - predicted)}$$

The average error in the original MW scale

7 How a prediction becomes a risk decision *(4 minutes)*

The model produces a continuous number in megawatts. Operators also need a simple status. GridPulse converts the absolute net-flow magnitude into three screening classes using thresholds calculated from the training period, not from the future test period.

Class	Rule in this run	Meaning
Normal	absolute flow < 3550.44 MW	Within the usual training envelope
Warning	3550.44 to 5941.40 MW	Needs attention and monitoring
Critical	absolute flow >= 5941.40 MW	High-magnitude condition needing investigation

This is a risk screen, not a physical safety certificate. The thresholds describe unusual flow magnitude in this dataset. They do not replace official line ratings or operating rules.

How to read the confusion matrix

A confusion matrix compares the actual class with the predicted class. Rows are actual conditions; columns are predicted conditions. Values on the diagonal are correct. Off-diagonal values show exactly which mistakes occurred.

Actual / Predicted	Normal	Warning	Critical
Normal	1490	58	0

Actual / Predicted	Normal	Warning	Critical
Warning	37	260	25
Critical	0	27	119

For example, Fusion GRU correctly identified 1,490 Normal cases, 260 Warning cases, and 119 Critical cases. It missed 27 Critical cases as Warning, which is more useful to inspect than a single overall accuracy number because missing a high-risk case is not the same as confusing two normal cases.

The main classification metrics

Metric	Plain meaning	Formula idea
Accuracy	Share of all classes predicted correctly.	correct predictions / all predictions
Precision	When the model raises a class, how often is it right?	true positives / predicted positives
Recall	Of the real cases in a class, how many did it find?	true positives / actual positives
F1	Balance between precision and recall.	2 x precision x recall / (precision + recall)
ROC AUC	How well a score separates positive and negative cases over thresholds.	area under true-positive-rate vs false-positive-rate
PR AUC	How useful the score is when positive cases are less common.	area under precision vs recall

8 How prescriptive rerouting works *(4 minutes)*

The prescriptive layer starts only after a forecast or observed condition indicates stress. It does not claim to operate the grid. It creates a transparent what-if scenario for a human reviewer.

The rerouting sequence

1. Identify the stressed node or link and record the reason: high predicted magnitude, high utilization, imbalance, or data-quality warning.
2. Build candidate paths through the canonical graph. A path must be connected from the source to the target.
3. Reject a candidate if it contains a Critical link, lacks enough headroom, or violates a hard guardrail.
4. Estimate the flow moved onto the candidate path and compute before-and-after screening values.
5. Rank feasible candidates by a combined score involving headroom, estimated loss, risk, distance, and data quality.
6. Show the best available option and any bottleneck, even when every option is imperfect. The user decides what to investigate.

Distance and power-loss proxy

Latitude and longitude are used to estimate the straight-line distance between country centroids. The result is intentionally labeled a proxy because the real conductor route and electrical parameters are not in the supplied data.

$$R = \rho \times L / A$$

Resistance estimate: resistivity × length / conductor area

$$S = |P| / \text{power factor};\ I = S / (\sqrt{3} \times V)$$

Apparent power and three-phase current estimate

$$P_loss = 3 \times I^2 \times R$$

Three-phase I-squared-R loss proxy

Assumption	Value in this run	Why it must be replaced for engineering use
Line voltage	400 kV	Actual voltage may differ by line and operating condition
Power factor	0.95	Actual reactive power is not provided
Resistivity	1.68e-8 ohm m	Conductor material and temperature are not provided
Cross-sectional area	2,000 mm2	Actual conductor bundle is not provided
Length	Haversine centroid distance	Actual route geometry is not a straight line

The project deliberately separates a useful scenario estimate from an engineering-grade calculation. A future version would require surveyed route length, conductor configuration, resistance at temperature, voltage, power factor, thermal ratings, and validation against a power-flow solver.

9 Results and what they mean *(4 minutes)*

The final comparison uses a common chronological holdout. RMSE and MAE measure continuous flow error in MW. Accuracy, precision, recall, and F1 measure the three risk classes. Inference time shows the cost of producing one prediction in the recorded experiment environment.

Model	RMSE MW	MAE MW	Accuracy	F1	Time ms
Persistence	580.3	375.8	91.4%	0.812	0.45
24h moving average	1363.4	947.1	81.3%	0.579	12.80
Linear trend	1604.8	1156.9	74.4%	0.512	78.02
Seasonal naive (24h)	1994.7	1310.5	75.4%	0.516	0.55
Fusion GRU	570.5	400.3	92.7%	0.856	34.73
DCRNN	634.0	436.2	92.5%	0.838	63.63

Research selection rule: the saved experiment chooses the lowest pooled RMSE on the common chronological holdout. Under that rule, Fusion GRU wins with RMSE 570.5 MW. Persistence remains the operational baseline because it is extremely fast and easy to audit, and it has lower MAE in this run. The two statements can both be true: one model is the best research forecast by RMSE, while a simpler rule remains attractive for operations.

DCRNN records RMSE 634.0 MW and F1 0.838, below Fusion GRU on this dataset. The result is not a failure of the idea; it is a boundary finding. The graph model was implemented and measured under the same protocol, but the available features and small topology do not provide enough evidence that diffusion improves the forecast here.

Error analysis: what worked and what failed

Case	Observed	Predicted	Interpretation
Success	SE at 2024-12-26 20:00: -5067.0 MW	-5067.2 MW	The model preserved the Warning class with almost no numerical error.

Case	Observed	Predicted	Interpretation
Failure	DE at 2024-12-28 14:00: 7484.5 MW	3986.7 MW	The sudden Critical jump was underestimated as Warning; this is an important edge case.

The right response to a failure is not to hide it. A reviewer should ask whether the event was caused by a sudden market change, missing external variables, a data gap, or a physical constraint absent from the dataset. Error analysis turns a score into a research question.

The nine most useful result visuals

- Dataset profile chart: shows the scale and shape of the source data.
- Preprocessing and feature catalog: shows what was cleaned and what was extracted.
- Training versus validation curves: checks learning, overfitting, and underfitting.
- Confusion matrix: shows which risk classes are confused.
- ROC and precision-recall curves: show threshold behavior, especially for rare Critical cases.
- Model comparison bars: compare RMSE, F1, accuracy, and inference time.
- Feature permutation and ablation chart: shows which inputs affect validation error.
- t-SNE embedding plot: shows whether hidden representations organize risk classes.
- Error-analysis examples: show concrete successes and failures instead of only averages.

10 How the software is integrated *(2 minutes)*

The project is organized as a small local application. The experiment runner creates evidence files, the backend exposes those files and operational calculations through API routes, and the browser renders an animated historical replay with charts and recommendations.

Layer	Technology	Use
Data and experiments	Python, pandas, NumPy	Read CSVs, align data, create features and artifacts
Machine learning	PyTorch	Train Fusion GRU and DCRNN checkpoints
Metrics and analysis	scikit-learn	Metrics, confusion matrices, ROC/PR, t-SNE
Figures	Matplotlib	Save light-background review figures
Backend	Python standard library HTTP server	Serve REST endpoints and the historical stream
Frontend	HTML, CSS, JavaScript	Animate the grid, charts, route plans, and explanations
Artifacts	JSON, CSV, PNG, PT checkpoints	Keep results inspectable and reproducible

The current frontend is intentionally lightweight HTML, CSS, and JavaScript rather than React. It keeps the project easy to run locally and makes the data flow visible. The backend exposes health, metadata, topology, snapshot, forecast, metrics, validation, physical-loss, route-plan, recommendations, what-if, and stream endpoints.

The live-looking values change because the browser advances through stored hourly observations and calls the backend for derived risk, forecast, loss, and routing information. It is a replay designed for demonstration, not live utility telemetry.

11 What is novel, and what are the limitations *(2 minutes)*

The practical novelty

The novelty is the integration and auditability of the workflow, not a claim of a brand-new neural layer. The project combines a canonical edge audit, data-quality-aware risk, evidence-driven model selection, explicit separation of proxy calculations from engineering facts, distance-to-loss bridging, stress-aware path search, and an explainable action queue.

- The model router does not assume DCRNN must win; it selects from evidence.
- The interface shows assumptions and failure modes instead of only showing a score.
- The rerouting mechanism exposes bottlenecks and infeasibility rather than hiding them.
- Every important result is saved under results/ before the presentation or notes are built.

Limitations to say openly

- The data is historical and the dashboard is a replay, not a live utility connector.
- The line-loss calculation is a distance-based three-phase proxy, not engineering-grade power-flow analysis.
- The dataset lacks actual line ratings, voltage, conductor details, reactive power, dispatch constraints, weather shocks, and market information.
- The stress labels are quantile-based screening labels, not official operating limits.
- Recommendations are for investigation and planning; they do not authorize switching or dispatch.
- Generalization requires testing on a documented benchmark or new time period and validating physical parameters.

12 One minute recap and viva questions *(2 minutes)*

Remember these eight points

- The project studies hourly cross-border power transfers across 12 countries and 24 canonical links.
- The dynamic CSV describes country conditions; the power-flow CSV describes interconnector movement.
- Preprocessing parses, validates, canonicalizes, aligns, scales, and splits the data chronologically.
- Feature extraction creates 11 signals: 4 local, 4 global, and 3 topology features.
- A 24-hour window becomes a 24 x 12 x 11 model input and the target is the next-hour 12-node flow vector.
- Fusion GRU and DCRNN are compared with simple baselines, so complexity must earn its place.
- The forecast becomes Normal, Warning, or Critical using training-only flow-magnitude thresholds.
- Rerouting is a guarded what-if recommendation based on headroom, risk, distance, proxy loss, and data quality.

Likely questions and short answers

Question	Answer
Why not use only accuracy?	Because accuracy can hide Critical-class mistakes. RMSE, MAE, precision, recall, F1, confusion matrices, and error examples give a fuller view.
Why is Fusion GRU the winner if Persistence is simpler?	Fusion GRU has the lowest RMSE in this run, but Persistence is faster, easier to audit, and has lower MAE. The right choice depends on the operational objective.
Why did DCRNN lag?	The graph is small and the available data lacks physical constraints and richer spatial signals. Graph context was tested rather than assumed to help.
Is the loss value real?	It is a distance-based scenario proxy using stated assumptions. Engineering-grade loss needs surveyed line and electrical parameters.
Does rerouting operate the grid?	No. It screens alternate paths and produces a human-review recommendation with guardrails.

Question	Answer
Where are the reproducible results?	Under results/: dataset profile, preprocessing summary, feature catalog, model comparison, curves, confusion matrices, ROC/PR, embeddings, importance, errors, figures, and checkpoints.

Where to look next in the project

Need to understand	Open this
Project purpose and run commands	README.md
Data and pipeline design	docs/SRS.md and backend/pipeline.py
Research gap and novelty	docs/RESEARCH_NOTES.md
Experiment implementation	backend/experiments.py
Dashboard behavior	docs/USER_MANUAL.md and frontend/app.js
Saved evidence	results/

Source in project: This guide is grounded in the generated results/ artifacts, the project README, the research notes, and the implemented pipeline and experiment code.